Unraveling the Wacker Oxidation Transition State for Anti-Markovnikov Product Formation

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Structure-Effect Analysis of Wacker-Type Catalysts: A Computational Approach

The selective formation of anti-Markovnikov products in alkene oxidation reactions presents an exciting challenge with wide-ranging applications in green chemistry and sustainable processes.[1,2] Traditional oxidation methods, such as the Tsuji-Wacker process, predominantly yield Markovnikov products, e.g. ketones from alkenes, due to the favored reaction pathway, limiting the catalytic transformations available for producing valuable chemical intermediates.[2] However, the development of novel catalytic strategies aimed at shifting product selectivity toward anti-Markovnikov products, e.g. aldehydes from alkenes, could revolutionize both industrial and academic chemical synthesis.[3] Achieving efficient anti-Markovnikov selectivity offers significant advantages in synthesizing aldehydes, which are key ingredients in fragrances, pharmaceuticals, fungicides, food additives, and polymer manufacturing.[2,4] This work seeks to unravel the transition state mechanisms governing anti-Markovnikov product formation through computational chemistry, as understanding these mechanisms is crucial for identifying the factors that dictate product selectivity and for overcoming the inherent preference for Markovnikov pathways in traditional oxidation processes. Building on our prior optimization of Wacker-type oxidation of 1-decene to n-decanal,[5] this study employs Density Functional Theory (DFT) to investigate the energetic profiles and geometries of transition states, particularly analyzing both conventional Wacker oxidation mechanisms and a novel pathway.

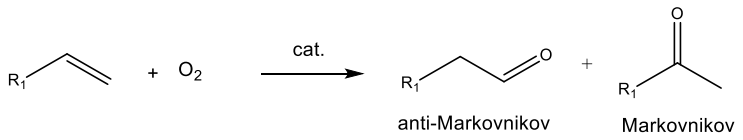

Figure 1: Direct oxidation of terminal alkenes to the anti-Markovnikov and Markovnikov products.[6]

All density functional theory (DFT) calculations were carried out using Gaussian 16. Geometry optimizations were carried out with the B3LYP method in the gas phase.

The 6-31G(d) basis set was applied for main group elements, while LANL2DZ and associated polarization functions were used for Pd. Additional polarization functions were added for Pd (ζ(f) = 1.472) , and frequency analysis was performed to confirm stationary points and compute thermodynamic corrections.[3,7] Key factors influencing selectivity, including catalyst structure, reaction intermediates, and solvent effects, were explored, providing insights for optimizing catalytic systems and promoting sustainable chemical synthesis. Two mechanisms were evaluated, both incorporating water as the nucleophile: one, similar to a *syn*-addition, used water directly bound as a ligand to the catalyst complex, while the other, the *anti*-addition, utilized atmospheric water.[8] The energy difference for the bound water mechanism is 34.32 kJ/mol, while the atmospheric water mechanism exhibites a larger difference of 51.77 kJ/mol, indicating a higher energy barrier for anti-Markovnikov product formation in the secound mechanism, while the Markovnikov route is favoured for both proposed mechanisms. These insights will further be applied in future investigations and to generate experimental data, aimed at improving the anti-Markovnikov selectivity.

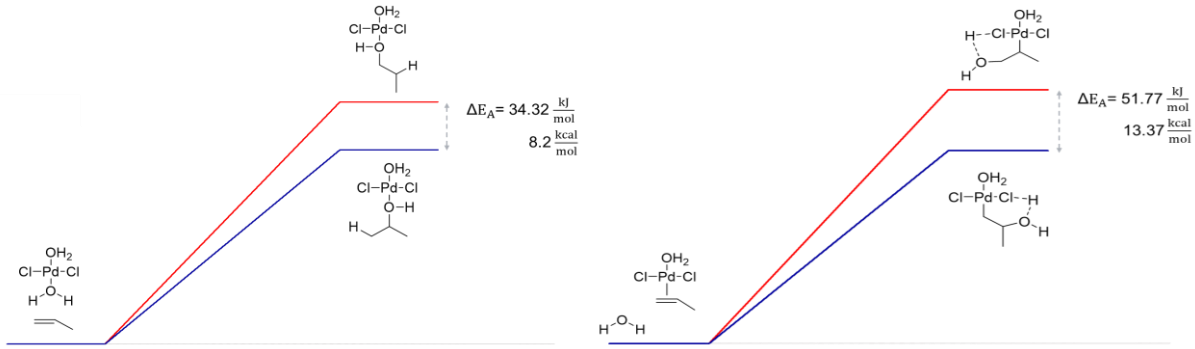

Figure 2: Reaction Pathways and Energy Barriers to the Transition State in Wacker Oxidation: Comparison of Water-Bound and Atmospheric Water Mechanisms for the Markovnikov and anti-Markovnikov products.

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